MEGA EXPO 2025

Problem Statement Title : Universal Smart Vehicle Safety & Monitoring

Track : Smart Transportation & Safety

PS Category : Both Hardware & Software

Team ID :

Team Name: Visiondrive

Universal smart vehicle safety & Monitoring system

PROBLEM:

Road accidents are increasing at an alarming rate and have become a major public concern causing loss of life injuries and property damage

PROPOSED SOLUTION:

- A multi-sensor safety module that detects accidents risky driving ,blind spots ,and vehicle condition.
- It reduces major road accidents by alerting drivers and emergency services instantly.
- The solutionis unique because it is vehicle - independent low cost ,plug-and-play,and works for cars,and 4-wheeler without modification.

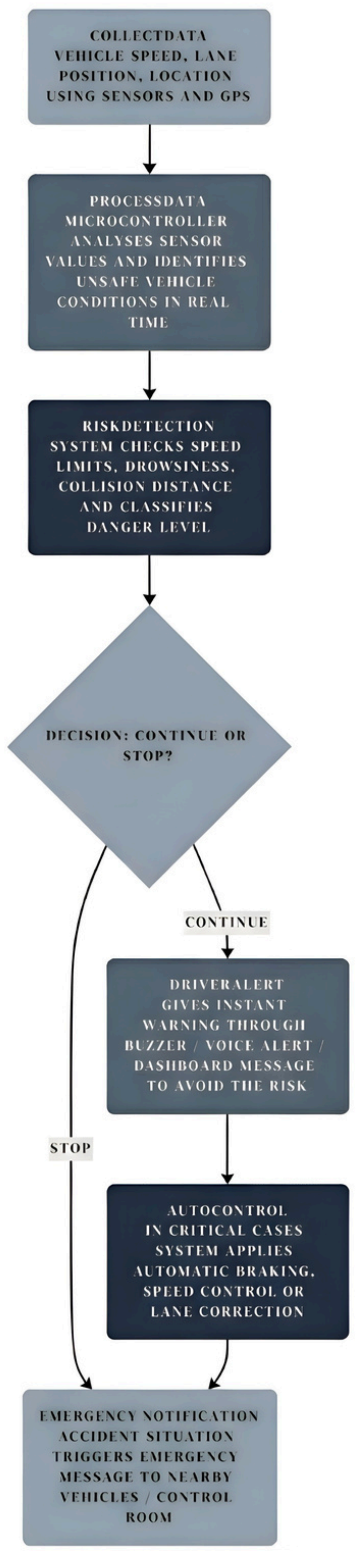
TECHNICAL APPROACH

Programming languages and Tools:

Arduino IDE (C / C ++), Python, Firebase, Arduino UNO / ESP32 Ultrasonic Sensor, IR Sensor, Speed Sensor, Accelerometer + Gyroscope (MPU6050), GPS Module, GSM / IOT Module, Battery.

Working Process:

1. **Collect Data** - Vehicle speed, lane position, location using sensors and GPS.
2. **Process Data** - Microcontroller analyses sensor values and identifies unsafe vehicle conditions in real-time.
3. **Risk Detection** - System checks speed limits, drowsiness, collision distance and classifies danger level.
4. **Driver Alert** - Gives instant warning through buzzer / voice alert / dashboard message to avoid the risk.
5. **Auto Control** - In critical cases system applies automatic braking, speed control or lane correction.
6. **Emergency Notification** - Accident situation triggers emergency message to nearby vehicles / control room.



TECHNICAL APPROACH

Programming languages and Tools:

Arduino IDE (C / C ++), Python, Firebase, Arduino UNO / ESP32 Ultrasonic Sensor, IR Sensor, Speed Sensor, Accelerometer + Gyroscope (MPU6050), GPS Module, GSM / IOT Module, Battery.

Working Process:

1. **Collect Data** - Vehicle speed, lane position, location using sensors and GPS.
2. **Process Data** - Microcontroller analyses sensor values and identifies unsafe vehicle conditions in real-time.
3. **Risk Detection** - System checks speed limits, drowsiness, collision distance and classifies danger level.
4. **Driver Alert** - Gives instant warning through buzzer / voice alert / dashboard message to avoid the risk.
5. **Auto Control** - In critical cases system applies automatic braking, speed control or lane correction.
6. **Emergency Notification** - Accident situation triggers emergency message to nearby vehicles / control room.

FEASIBILITY AND VIABILITY

Feasibility:

Technically feasible -

- Utilizes proven automotive sensors and widely available embedded systems.

Operationally feasible-

- Can be seamlessly integrated into existing vehicle platforms with minimal modificatin.

Viability:

- The solution is practically implementable using commercially available hardware and software.
- Low integration complexity allows adoption in existing and new vehicles.
- Cost-benifit ratio is favourable

Challenges:

- Integration complexity varies across different vehicle types and models.
- Costs of sensors and connectivity may limit large-scale adoption initiallly.

Strategies:

- Deploy real-time alert mechanism for immidiate driver response.
- Integrate multi-source vehicle and road data for reliable predictions.
- Countinuously optimises system accuracy through perodic models.

FEASIBILITY AND VIABILITY

FEASIBILITY

Technically feasible

Utilizes proven automotive sensors and widely available embedded systems

Operationally Feasible

Can be seamlessly integrated into existing vehicle platforms with minimal modification

VIABILITY

- The solution is practically implementable using commercially available hardware and software.
- Low integration complexity allows adoption in existing and new vehicles.
- Cost-benifit ratio is favourable

CHALLENGES

- Integration complexity varies across different vehicle types and models.
- Costs of sensors and connectivity may limit large-scale adoption initiallly.

STRATEGIES

- Deploy real-time alert mechanism for immidiate driver response.
- Integrate multi-source vehicle and road data for reliable predictions.
- Countinuously optimises system accuracy through perodic models.

IMPACT AND BENEFITS

Prevents Road Accidents

Potential Impacts :

- **Enhances road safety** by predicting collisions and reducing human errors in driving.
- **Improves public transportation safety** for 4 - wheelers, buses and all commercial vehicles.
- **Reduces accident rate** on highways and urban roads by real- time alerts.
- **Minimizes traffic congestion** through smart monitoring and preventive actions.

Benefits :

- **Supports emergency response** with quick information about risky zones.
- **Saves economic loss** caused by vehicle damages, medical expenses and traffic disruption.
- **Protects human life** by preventing accidents before they occur.

RESEARCH AND REFERENCES

Research :

https://doi.org/10.1016/j.procs.2018.04.060

https://www.analog.com/en/resources/analog-dialogue/articles/electric-vehicle-warning-sound-system.html

Image reference :

https://g.co/gemini/share/f96b43cd7360

Hardware & Software Components Required

Arduino IDE (C/C++), Python, Firebase, Arduino UNO/ESP32 Ultrasonic Sensor, IR Sensor, Speed Sensor, Accelerometer Gyroscope (MPU6050), GPS Module, GSM/IOT Module, Battery.